Mute Heart: IoT-Enabled Wearable Gesture Recognition Glove for Sinhala Sign Language Voice Translation

Final IoT Project Report

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Abstract

Communication barriers between hearing-impaired individuals and the general public remain a significant societal challenge, particularly in contexts where Sri Lankan Sign Language (SSL) is not widely understood. This project introduces 'Mute Heart', an IoT-enabled wearable gesture recognition glove designed to translate selected Sinhala sign language gestures into audible speech output. The system uses an ESP32 microcontroller to process data collected from five rotary potentiometers attached to the fingers and an MPU6050 motion sensor that captures hand movement. The processed gesture is mapped to predefined Sinhala words and played through a DFPlayer Mini audio module connected to a speaker.

The prototype is constructed using a glove frame supported by wooden sticks for mechanical stability. Programming was implemented using Arduino IDE (C/C++) and Python via PyCharm for supplementary processing. The system currently demonstrates six Sinhala vowel gestures: 'අ (Ayanna)', 'ආ (Aayanna)', 'ඇ (Aaeyanne)', 'එ (Eyanna)', 'ඒ (Eeayanna)', and 'උ (Uyanna)'. The device is powered through a USB connection to a computer during testing. Although the current implementation uses rule-based gesture detection, future versions may integrate machine learning using Raspberry Pi for improved gesture recognition accuracy.

Acknowledgement

We would like to express our sincere gratitude to our lecturer, Mr. Bathiya Seneviratne, for his guidance, support, and valuable feedback throughout the development of this project. His insights greatly contributed to the successful completion of the 'Mute Heart' IoT gesture recognition system.

We also thank the National Institute of Business Management (NIBM) for providing the resources, laboratory facilities, and academic environment required to complete this project. Finally, we appreciate the dedication and teamwork of all group members whose collaboration made this project possible.

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Introduction

Human communication is fundamental for social interaction and access to services. However, individuals with hearing and speech impairments rely heavily on sign language for communication. Sri Lankan Sign Language (SSL) provides a structured communication system for the deaf community, but the majority of the population is not fluent in this language, which creates communication barriers.

Technological advancements in Internet of Things (IoT) and wearable devices offer new opportunities to bridge this gap. Traditional gesture recognition systems often rely on camera-based vision techniques that require controlled environments and high computational power. In contrast, sensor-based wearable systems are more portable, reliable, and cost-effective.

This project presents 'Mute Heart', a wearable gesture recognition glove capable of converting Sinhala sign gestures into audible Sinhala speech. By combining potentiometer-based finger angle detection, motion sensing using an MPU6050 sensor, and embedded processing with the ESP32 microcontroller, the system can detect specific hand gestures and translate them into spoken words using a DFPlayer Mini audio module.

Methodology

The system follows a sensor-driven embedded architecture consisting of gesture sensing, signal processing, gesture classification, and voice output generation stages.

Sensing Mechanism: Five rotary potentiometers are attached to the fingers of a glove to detect angular movement. When the user bends a finger, the potentiometer rotates, producing an analog voltage corresponding to the finger angle. An MPU6050 inertial measurement unit is used to capture motion and orientation data of the hand.

Processing Unit: The ESP32 microcontroller collects analog signals from the potentiometers and motion data from the MPU6050 sensor. These signals are processed using rule-based threshold mapping to identify the performed gesture.

Voice Output: Once a gesture is recognized, the ESP32 sends a command to the DFPlayer Mini module, which plays a pre-recorded Sinhala audio file through a connected speaker.

Mechanical Structure: The prototype uses wooden sticks attached to the glove to stabilize sensors and maintain proper alignment with finger joints.

Software Development: Programming was implemented using Arduino IDE (C/C++) for microcontroller control and Python using PyCharm for additional experimentation and data handling.

Methodology – Joystick Glove for Gesture Control

To extend the functionality of the system, an additional **gesture control glove using a joystick module** was developed. This glove allows the user to perform hand movements that can be detected and interpreted as specific gestures.

Hardware Setup

The joystick module was attached to the glove so that it could move according to the user's hand movements. The joystick contains two potentiometers that measure movement in the **X-axis and Y-axis directions**. These analog values were connected to the **ESP32 microcontroller**, which reads the position of the joystick and converts it into digital signals.

The main components used in this part include:

- **Joystick Module** – Detects directional movement (X and Y axis)
- **ESP32 Microcontroller** – Processes joystick signals
- **Jumper Wires** – Connect components
- **USB Power Supply** – Provides power through the computer

Data Processing

The ESP32 continuously reads the analog values from the joystick module. Based on the movement of the joystick, the system determines the direction of the hand movement such as **left, right, up, or down**.

These directional inputs are then mapped to predefined gestures or commands in the program.

Gesture Recognition

When the user moves the joystick in a specific direction, the ESP32 detects the corresponding analog value range and identifies the intended gesture. The program then processes the gesture and sends the result through the serial monitor or triggers the output of the system.

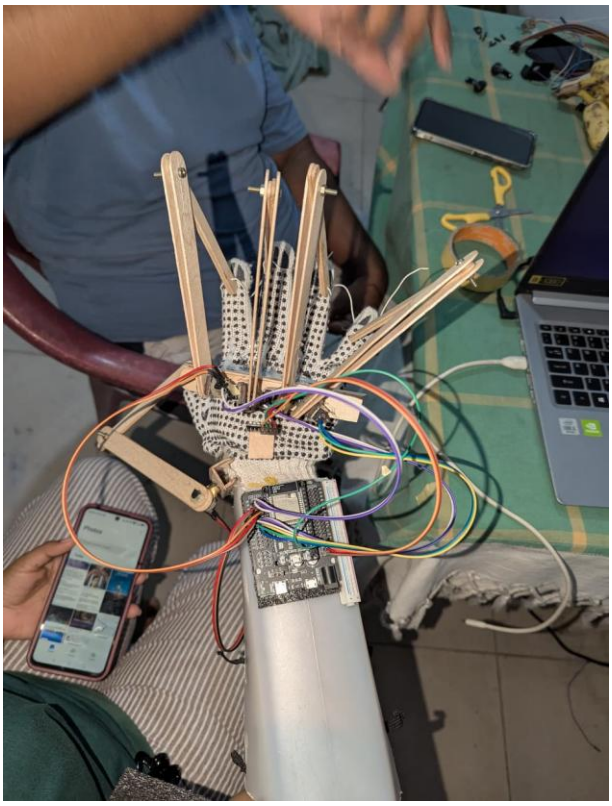
System Integration

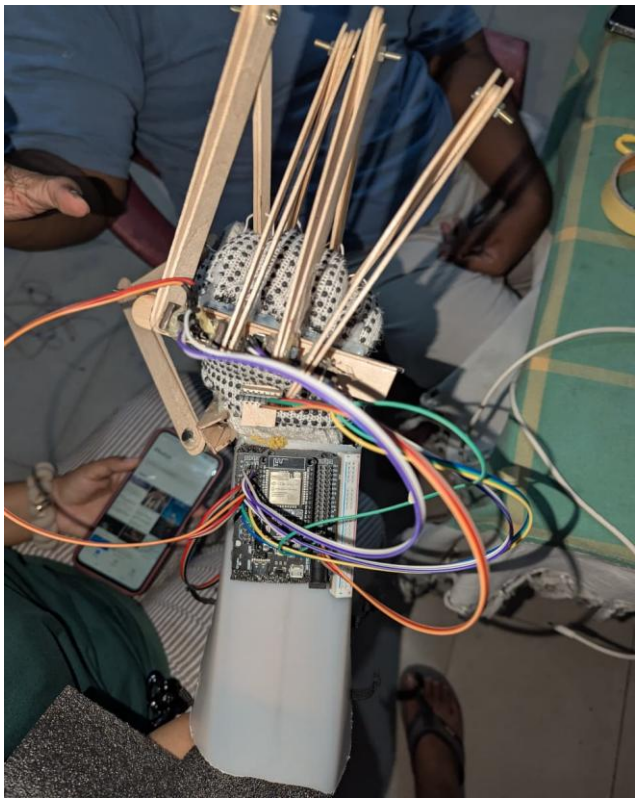
The joystick glove works as an additional input method alongside the flex sensor glove. This allows the system to demonstrate multiple gesture input techniques and improves the flexibility of the gesture recognition system.

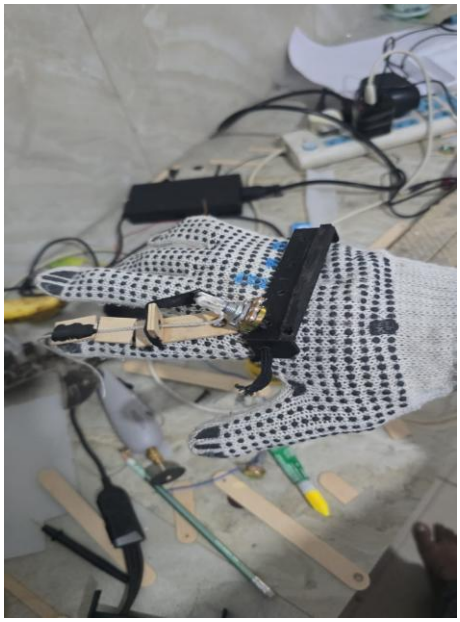
Outcome

The joystick glove successfully demonstrated an alternative method for gesture detection. By using directional movement instead of finger bending, the system could recognize gestures based on hand motion and direction.

Prototype Smart Glove (Actual Implementation)







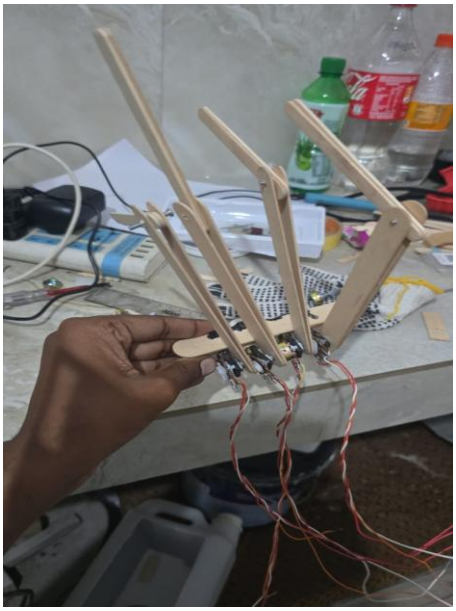


Figure: Developed smart glove prototype using wooden sticks and potentiometers.

Selected Sinhala Sign Gestures Used in the System

Figure: Sinhala sign language gesture references used to identify 6 selected words.

Z

Discussion

The developed prototype successfully demonstrates the feasibility of translating selected Sinhala sign gestures into audible speech using an embedded IoT device. The use of potentiometers allowed accurate detection of finger bending angles, while the MPU6050 sensor helped identify hand movement and orientation.

The wooden mechanical structure provided a simple and low-cost alternative to 3D-printed components, allowing rapid prototype development. However, manual assembly introduced minor alignment variations that could affect sensor accuracy. The USB power supply ensured stable operation during testing, although a portable battery solution would be necessary for real-world use.

Overall, the system achieved reliable detection of the five implemented gestures and successfully produced the corresponding Sinhala speech output.

Future Implementations

Several improvements can be introduced in future versions of the system. Machine learning algorithms could replace rule-based gesture recognition to increase accuracy and support a larger vocabulary of gestures. Integration with a Raspberry Pi device could provide additional computational power for implementing neural network models.

Further improvements include designing a fully 3D-printed glove frame for better mechanical alignment, integrating a rechargeable battery for portability, and developing a mobile application to display recognized words in real time. Expanding the gesture dataset to support the full Sinhala sign language alphabet would also significantly enhance the usability of the device.

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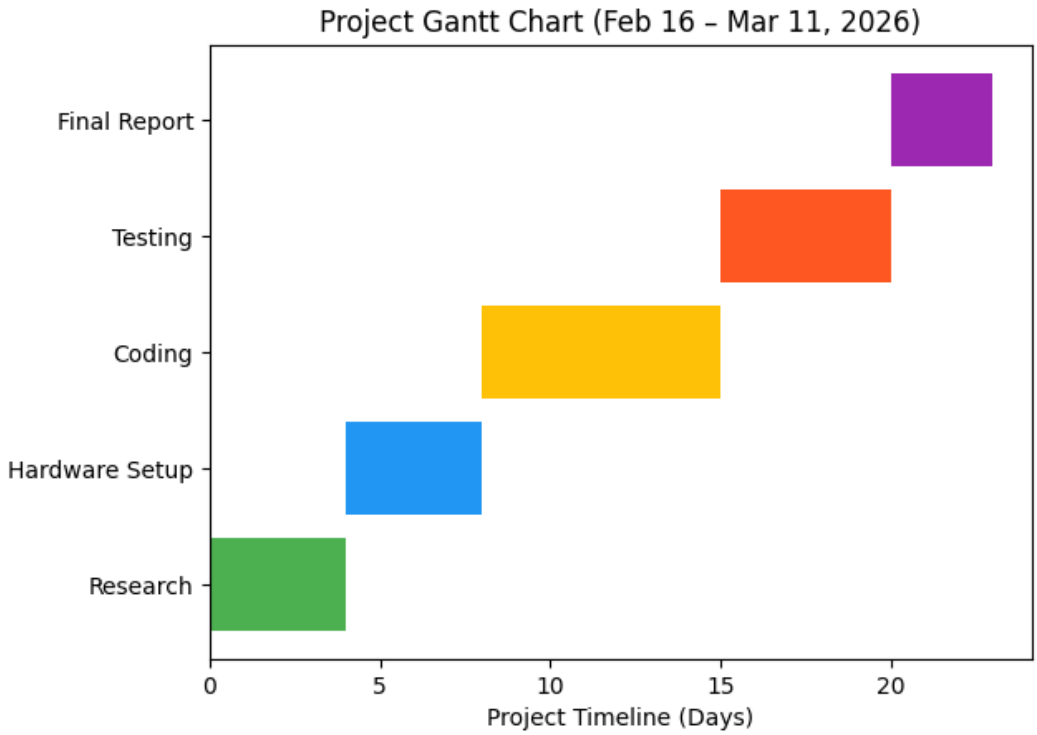
Gantt Chart



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